THE UNIVERSITY OF CHICAGO
DEPARTMENT OF ECONOMICS
ECON 20000 - ELEMENTS OF ECONOMIC ANALYSIS I
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Problem Set 3

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Instructions. The problem set is due on Tuesday Nov, 21*st* at the *beginning* of the lecture, in paper copy. You should work in groups of $3-5$ students and turn in one problem set per group.

1 Endowment economy with Cobb-Douglas

Suppose that a consumer utility function is:

$$u(x,y) = x^{\frac{1}{5}}y^{\frac{4}{5}}$$

and that her endowment is given by $(\omega_x, \omega_y) = (10, 10)$. Let $p_y = 10$ throughout the problem.

a. Suppose $p_x = 10$. Solve for the optimal consumption bundle of the consumer.

b. Now suppose that p_x drops to 5. What is the change in demand for good x? Calculate the different components of this effect. Is the consumer better off at this price? Explain.

c. For which values of p_x will the consumer be a net supplier of x?

d. For which value of p_x is the consumer's utility lowest? Explain the intuition of the result and use a diagram to describe it.

2 Endowment economy with perfect substitutes

Suppose that a consumer's preferences can be represented by the utility function:

$$u(x,y) = 2x + y$$

Let the consumer endowment be $(\omega_x, \omega_y) = (100, 200)$ and let market prices be $p_x = 5$ and $p_y = 2$.

a. Draw the consumer's budget line and label the initial endowment by E.

b. What are the consumer's gross and net demands? Label the consumer's chosen bundle by O in your plot.